

Computer vision: Then and now

In his whitepaper titled 'Everything You Ever Wanted To Know About Computer Vision,' Ilija Mihajlovic, an iOS developer who is passionate about machine learning (ML), explains how ML and deep learning have simplified computer vision. Before deep learning came into the picture, computer vision did not have as much scope as it does today, mainly because a lot of manual coding was required, such as creating a database, annotating images, capturing new images, etc.

"With machine learning, developers no longer needed to manually code every single rule into their vision applications. Instead, they program features or smaller applications that could detect specific patterns in images," writes Mihajlovic. Today, we use deep learning for computer vision, and all of the work is done by neural networks. When we provide neural networks with lots and lots of labeled images of a particular object, they are able to identify patterns in them.

Hardware for computer vision

Capturing data is an important aspect of computer vision applications, and hardware platforms play a major role here. The main components required for computer vision applications are cameras and image sensors for capturing images, I/Os, a communication interface, and a processing unit. Imaging sensors present in cameras are characterised by their resolution (number of pixels), speed (frames per second), and number of colours.

Depending on the type of computer vision application, the processing unit might vary from CPUs to embedded boards. Here are a few examples:

- CPUs
- GPUs
- Heterogenous systems (CPUs +GPUs+others)
- FPGAs
- ASICs
- Microcontroller boards

Computer vision at work!

For the 2018 Masters Golf event, IBM employed computer vision to produce My Moments. IBM Watson was given hundreds of hours of film, after which it was able to recognise the sights (and sounds) of key scenes. These pivotal moments were chosen and distributed to viewers as individual highlight clips!

- Embedded systems (smart cameras)

So where do development boards stand?

Traditional computer vision systems used a camera connected to a general-purpose computer, which used to run the application. In a smart camera though, the whole system can be integrated, right from capturing the image to processing to the final application. Today, the distinction between smart cameras and development boards has slowly started to fade, since many development boards have inbuilt cameras and are tailored for computer vision applications.

With more and more applications being deployed on edge, it makes more sense to deploy such devices on development boards. Here are some critical features that you need to

consider while selecting a development board for your computer vision application:

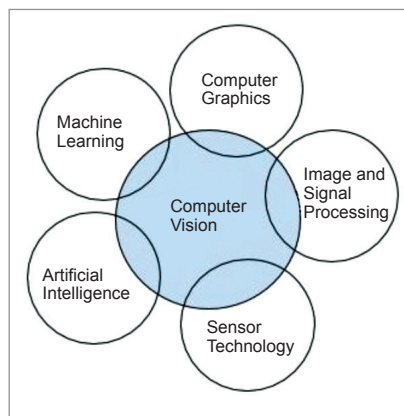
- High processing speed
- Edge processing ability
- Faster access to memory
- Low power consumption
- Portable
- Inexpensive

While there are embedded platforms specifically for AI like NVIDIA Jetson and Raspberry Pi Compute, AI accelerators like Google TPU, SOMs/ SBCs containing high-quality imaging sensors, we will focus on a few boards that were designed for computer vision and compare some of their features.

Current trends in computer vision

Computer vision on edge: To successfully deploy computer vision on edge, neural network design must be integrated into embedded platforms. Then, there are also difficult design decisions to be made around power consumption, cost, accuracy, and flexibility. Overall, for bandwidth, speed, and security reasons, edge computing is critical to computer vision processes. Still, hackers could take advantage of emerging security flaws in artificial intelligence. Designers must take this into consideration too.

However, there are some challenges when it comes to vision AI on edge, such as involving all edge endpoints can lead to complex device architectures. This makes



Fields that drive computer vision (Credit: IFA)

Components of computer vision

Hardware: Cameras/Imaging sensors

Software: Machine learning models for processing the images

Hardware: Processing units or conditional logic to automate application-specific use cases